

Abhimanyu Singh

Programmer Analyst | AI/ML Developer | Data Scientist

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 Github

PROFILE

Generative AI Developer with 3.8+ years of experience designing, developing, and deploying end-to-end GenAI pipelines, Natural Language Processing (NLP) applications, Computer Vision systems, and Machine Learning solutions. Skilled in Python, PyTorch, Transformers, and Google Cloud Platform (GCP). Proven track record of improving automation workflows, and delivering production-ready solutions for enterprise clients including Pfizer and Xerox. Holds a B.Tech in Computer Science.

PROFESSIONAL EXPERIENCE

- Programmer Analyst, Cognizant Technology Solutions**02/2025 – Present
Noida, Uttar Pradesh
- Automated helpdesk operations and ticket resolution workflows using Generative AI.
 - Designed intelligent solutions reducing manual effort by 80% with an accuracy of over 90%.
 - Integrated Databases, Knowledge Graphs, and Langchain/LangGraph to build conversational AI assistants.
- Software Engineer, HCLTech**07/2022 – 02/2025
Noida UP, India
- Developed AI-based document classification system using OCR and Transformer-based classification models.
 - Experimented with LLMs for summarization, text generation, and Retrieval-Augmented Generation (RAG).
 - Gained hands-on experience working with open-source vision-based LLMs.

EDUCATION

- Bachelor of Technology - Computer Science, Rajasthan Technical University**07/2018 – 07/2022
Percentage: 84%Jaipur, Rajasthan, India
- Higher Secondary, Brij Bal Sr Secondary School**03/2017 – 07/2018
Percentage: 74%Jaipur, Rajasthan, India

SKILLS

- Programming & Frameworks**
Python, NumPy, pandas, Scikit-learn, Tensorflow, PyTorch, Transformers, Keras, OpenCV, Flask, FastAPI, Streamlit, MCP
- AI/ML & Deep Learning**
Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, CNN, RNN, ResNet, OCR, CUDA
- NLP & Generative AI**
LLMS, BERT, ROBERTa, NLTK, Tokenization, TF-IDF, Embeddings, Prompt Engineering, LangChain, LangGraph LlamaIndex, RAG, Vector DBs, Semantic Search, LLMOps, Agentic AI
- Cloud & DevOps**
GCP (Vertex AI, Dialogflow, AI Studio, Cloud Run, Artifact Registry), AWS, Azure, Docker, Git, CI/CD

PROJECTS

Generative AI Service Now Agent

- Designed a Generative AI-powered Service Now assistant to automate application-level troubleshooting using log analysis.
- Integrated langgraph + FastAPI for intent recognition and multi-turn conversations.
- Implemented short-term and cross-session memory using langgraph.
- Reduced manual support by 50% and improved ticket resolution SLA compliance by 40%.
- Tech Stack: Langgraph, ChromaDB, Python, FastAPI

Hybrid Retrieval-Augmented Generation (RAG) knowledge Engine

- Built a custom Hybrid RAG pipeline for enterprise knowledge management using Langchain & ChromaDB.
- Enhanced accuracy of LLM response through document chunking, embeddings, and hybrid retriever.
- Supported multi-model inputs (text, PDFs scanned documents) for scalable enterprise deployments.
- Tech Stack: Langchain, Chromadb, Azure OpenAI, FastAPI

Intelligent Document Processing (Document AI)

- Developed an end-to-end Document Intelligence system combining OCR+NLP+RAG pipelines for structured data extraction.
- Applied PaddleOCR & Tesseract for text recognition, Spacy/NLTK for NER, and Langchain + FAISS for contextual search.
- Integrated Responsible AI guardrails:
- PII masking & redaction for sensitive data.
- Input validation & response moderation to handle toxicity, bias, and prompt injection attacks.
- Hallucination control based on document context similarity score.
- Achieved 95% entity recognition accuracy and automated data validation workflows, reducing manual effort by 70%.
- Tech Stack: Python, PaddleOCR, Spacy, Langchain, Guardrails, FAISS, SQL, HuggingFace Transformer models.

CERTIFICATES

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| • Introduction to Large Language Models | • Developing AI Applications with Python and Flask | • Complete Agentic AI Bootcamp with Langgraph and Langchain |
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